

Modeling Depression: Extensions, Evaluation, and Individual Recovery

Aidan Batista

Duke University

Mathematical Modeling Seminar

April 26, 2026

Abstract

Major depressive disorder affects hundreds of millions of people worldwide, yet the processes governing episode onset, persistence, and recovery remain poorly characterized at the individual level. Mathematical models provide a rigorous alternative to biological measurement, as they extract clinically interpretable parameters directly from longitudinal self-reported data. This paper develops and evaluates three such frameworks. A discrete-time Markov chain, extended to a Hidden Markov Model, accounts for observation noise in population hazard data. A Gamma-distributed autoregressive model, extended with a structural break, detects treatment-induced shifts in symptom persistence and baseline severity. An Ornstein-Uhlenbeck random mood model, extended via the Kalman filter, moves from population-level recovery curves to individual-level latent mood inference. Two datasets ground the analysis: NEMESIS, a Dutch psychiatric epidemiologic survey providing population time-to-event data across 250 MDD patients, and [Kossakowski et al. \[2017\]](#), an intensive 238-day experience sampling record from a single MDD patient. The HMM identifies distinct high- and low-recovery phases in population hazard dynamics. The AR(1) model captures a persistence of $\hat{a} = 0.30$, with simulated treatment reducing stationary severity by 91%. The extended OU model recovers an individual relaxation time of $T = 7$ days against a population estimate of $\langle T \rangle = 365$ days, revealing a sharp disconnect between daily affect fluctuations and clinical episode timescales.

Contents

1	Introduction	3
2	Background and Clinical Context	4
2.1	Defining Major Depressive Disorder	4
2.2	Self-Scoring Metrics as Sufficient Data	5
3	Data	6
3.1	Kossakowski et al. (2017) — Individual ESM Data	6
3.2	van der Werf et al. (2006) — NEMESIS Population Data	7
4	Models	8
4.1	Markov Model and Hidden Markov Model Extension	9
4.1.1	Markov Model	9
4.1.2	Hidden Markov Model Extension	10
4.2	Autoregressive Model and Treatment Extension	12
4.2.1	AR(1) Gamma Model	12
4.2.2	Treatment Extension	13
4.3	Ornstein-Uhlenbeck Random Mood Model and Individual Extension	16
4.3.1	OU Model Formulation	16
4.3.2	Individual Extension via Kalman Filter	18
5	Synthesis and Model Comparison	20
6	Discussion	23

1 Introduction

Major depressive disorder (MDD) is one of the most prevalent and disabling psychiatric conditions worldwide. Despite decades of clinical research, the mechanisms underlying the onset, persistence, and recovery from depressive episodes remain poorly understood at the individual level. Mathematical models offer a complementary approach: rather than targeting the biological substrate of depression directly, they describe the observable dynamics of mood over time, providing quantitative frameworks for understanding relapse risk, treatment response, and individual variation in recovery.

Direct biological measurement of depression remains impractical for longitudinal modeling. Neurotransmitter levels require invasive or expensive assays, individual neurobiological mechanisms are largely unknown, and most biological findings represent correlational group averages rather than individual-level signals. Self-report severity scores fill this gap. The PHQ-9 achieves 88% sensitivity and 88% specificity against structured clinical interview [Kroenke et al., 2001], is brief enough to administer at every visit, and produces time series with well-characterized distributional structure. That structure, positively skewed and Gamma-distributed, is not a measurement artifact to be corrected but a statistical signal that directly constrains model choice. Noise in self-report data is quantifiable; it belongs inside the model, not outside it.

The mathematical modeling literature that I reference has taken two broad approaches. Cochran et al. [2017] compare eight models of mood in bipolar disorder, ranging from biological rhythms oscillators to discrete-time Markov chains and autoregressive models, evaluating each against simulated mood time courses via survival functions, probability density functions, and spectral density. Their central finding is that visual inspection of time courses alone cannot distinguish between fundamentally different model assumptions, motivating the need for rigorous statistical evaluation. Van der Werf et al. [2006] take a different approach, fitting an Ornstein-Uhlenbeck random mood model to population time-to-event data from

250 MDD patients in the NEMESIS cohort. They recover interpretable parameters, a stimulus spacing of $D = 120$ days and a relaxation time of $\langle T \rangle = 365$ days, and demonstrate that the 14-day DSM inclusion criterion artificially suppresses early hazard rates. Both papers treat self-report data as the primary signal and extract meaningful structure from it.

This paper builds directly on both references. I extend three baseline models from [Cochran et al. \[2017\]](#), the Markov chain, the Gamma AR(1), and the OU random mood model, each with a component that addresses a specific limitation of the original. The Markov chain is extended to a Hidden Markov Model to replace hard state thresholds with probabilistic state inference. The AR(1) is extended with a structural break to detect treatment-induced shifts in symptom dynamics. The OU model is extended from population-level hazard fitting to individual-level latent mood inference via the Kalman filter. Two datasets ground the analysis: the NEMESIS population cohort from [van der Werf et al. \[2006\]](#) and an intensive 238-day experience sampling record from a single MDD patient [[Kossakowski et al., 2017](#)]. The individual-level OU model extension is the paper’s central contribution, bridging population recovery dynamics and personalized parameter estimation. Section 2 establishes the clinical context and argues for the sufficiency of self-report data. Section 3 describes the datasets. Section 4 develops each model and its extension. Section 5 compares the three frameworks. Section 6 discusses limitations and directions for future work.

2 Background and Clinical Context

2.1 Defining Major Depressive Disorder

According to the fifth edition of the Diagnostic and Statistical Manual of Mental Disorders (DSM-5) [[American Psychiatric Association, 2013](#)], a diagnosis of MDD requires the presence of five or more of nine specified symptoms over a two-week period, with at least one symptom being either depressed mood or markedly diminished interest or pleasure in activities. These

symptoms include changes in sleep, appetite, energy, concentration, psychomotor functioning, feelings of worthlessness, and recurrent thoughts of death. The categorical nature of this diagnostic system (that a patient either meets criteria or does not) stands in tension with the continuously varying nature of mood over time.

Depression carries a genetic heritability of approximately 37%, with risk distributed across many genes rather than concentrated in a few [[American Psychiatric Association, 2013](#)]. Neuroimaging studies implicate altered function in the amygdala (which processes emotional salience), the prefrontal cortex (which regulates cognitive control over affect), striatal circuits (which govern reward and motivation), and the hippocampus (which encodes contextual memory). Dysregulation of serotonin, dopamine, and norepinephrine systems is widely cited, and most pharmacological treatments target one or more of these pathways. These findings, however, are correlational group averages: individual neurobiological mechanisms remain largely unknown, neurotransmitter levels cannot be measured directly without invasive procedures, and no validated biological marker exists that tracks depression severity longitudinally in clinical practice.

2.2 Self-Scoring Metrics as Sufficient Data

Given the difficulty of direct biological measurement, mathematical models of depression are necessarily grounded in self-report severity scores. Two instruments are widely used. The Patient Health Questionnaire-9 (PHQ-9) is a patient-administered scale rating nine DSM symptom criteria on a 0–3 frequency scale, yielding a total score of 0–27 [[Kroenke et al., 2001](#)]. Severity is conventionally categorized as minimal (0–4), mild (5–9), moderate (10–14), moderately severe (15–19), and severe (20–27). The PHQ-9 achieves a sensitivity of 88% and specificity of 88% for MDD diagnosis against structured clinical interview, and is sufficiently brief to be administered at every clinical visit. The Montgomery-Åsberg Depression Rating Scale (MADRS) is a clinician-administered instrument rating ten items on a 0–6 scale, providing a more standardized but less frequently collected assessment.

A natural concern is whether self-report data is reliable enough to support mathematical modeling. I argue that it is, for three reasons. First, clinical validation studies confirm that PHQ-9 scores are statistically robust predictors of clinical outcomes. Second, the distributional structure of self-report scores (positively skewed and well-approximated by a Gamma distribution [Cochran et al., 2017]) is itself a scientifically meaningful signal that directly motivates modeling choices. Third, both primary references for this paper [van der Werf et al., 2006, Cochran et al., 2017] build quantitative models entirely from self-report time series and obtain clinically meaningful results. Noise in self-report measurements is quantifiable and modelable; it is not a reason to reject the data, but rather to build models that explicitly account for it.

Category	Score Range	Clinical Action
Minimal	0–4	None indicated
Mild	5–9	Watchful waiting
Moderate	10–14	Treatment plan
Moderately Severe	15–19	Active treatment
Severe	20–27	Immediate intervention

Figure 1: PHQ-9 severity categories and associated clinical thresholds.

3 Data

Two publicly available datasets are used in this paper, each serving a distinct role in the modeling framework.

3.1 Kossakowski et al. (2017) — Individual ESM Data

The first dataset is an intensive longitudinal study of a single participant with a history of MDD, monitored over 238 consecutive days using the experience sampling method (ESM) [Kossakowski et al., 2017]. Up to ten momentary mood assessments were collected per day via a handheld device, yielding approximately 1,476 observations across 238 days. The primary

variable used in this paper is the daily mean of the `mood_down` item, a momentary rating of “feeling down” on a -3 to $+3$ scale. This is aggregated to 238 daily observations and used for all individual-level modeling. A composite depression severity score (`dep`) derived from 13 SCL-90-R items is available at weekly resolution (187 observations) and used for state classification in the Markov section.

The dataset spans five study phases. Phases 1–3 correspond to relatively stable, lower-severity periods, while phases 4–5 correspond to increasing depressive symptomatology. The study features a notable design: the participant was followed through a period of relative stability into a worsening phase, making it well-suited for studying how individual mood dynamics evolve over time. The dataset is freely available at <https://osf.io/j4fg8> under a CC-BY license.

It is important to note that this dataset does not use PHQ-9 or MADRS directly. The ESM affect items and SCL-90-R composite score are analogous self-report instruments capturing the same depressive constructs at higher temporal resolution than weekly clinical assessments.

3.2 van der Werf et al. (2006) — NEMESIS Population Data

The second dataset is derived from the Netherlands Mental Health Survey and Incidence Study (NEMESIS) [van der Werf et al., 2006], a prospective psychiatric epidemiologic survey of the Dutch population aged 18–64 conducted in three waves (1996, 1997, 1999). From this survey, 250 subjects with a new major depressive episode between the first and second waves were identified. Diagnosis was made using the Composite International Diagnostic Interview (CIDI) under DSM-III-R criteria. Episode duration was assessed at 3-month intervals via the Life Chart Interview and retrospectively discretized into 6-week inspection windows.

The data used in this paper are the population-level time-to-event summaries from Table 1 of van der Werf et al. [2006]: the number of subjects still depressed (n_i), number recovered

(r_i) , and number censored at each inspection interval. From these, empirical hazard rates $\hat{h}_i = r_i/n_i$ and Kaplan-Meier survival probabilities $\hat{S}(t_k) = \prod_{i=1}^k (1 - \hat{h}_i)$ are computed directly. These 18 data points constitute the population-level signal that the Markov and OU models are fitted to.

The two datasets are complementary by design. NEMESIS captures episode-level recovery dynamics across 250 patients over 24 months, providing the statistical power needed to estimate population hazard rates and survival probabilities but offering no window into any individual’s day-to-day experience. Kossakowski provides the opposite: 238 days of daily mood measurements from a single person, rich enough to fit individual-level parameters but insufficient for population inference. Neither dataset alone supports the full modeling agenda of this paper. Together, they allow the same mathematical framework to operate at two scales simultaneously, fitting population parameters from NEMESIS and individual parameters from Kossakowski, and the contrast between the two sets of estimates is itself one of the paper’s central findings.

4 Models

I consider three classes of mathematical models for depression dynamics, each extending a baseline formulation from Cochran et al. [2017] with a new component motivated by the limitations of the original. For each model, I describe the mathematical formulation, the fitting procedure, the extension, and the results on data. All code used in this analysis is available at <https://colab.research.google.com/drive/1H7v3pUE7ITIL81V7wcUgVjK1PA1V4dTa?usp=sharing>.

4.1 Markov Model and Hidden Markov Model Extension

4.1.1 Markov Model

The discrete-time Markov chain (DTMC) treats depression as a categorical state evolving over time. Let $S_t \in \{D, R\}$ denote the state of the population at time t , where D denotes the depressed phase and R the remitted phase. The model is characterized by the memoryless (Markov) property:

$$\Pr(S_{t+1} = j \mid S_t = i, S_{t-1}, \dots, S_0) = \Pr(S_{t+1} = j \mid S_t = i) \quad (1)$$

The transition probabilities are collected into a 2×2 matrix:

$$P = \begin{pmatrix} P_{DD} & P_{DR} \\ P_{RD} & P_{RR} \end{pmatrix} \quad (2)$$

where $P_{ij} = \Pr(S_{t+1} = j \mid S_t = i)$. The probability distribution over states at time t evolves as $\boldsymbol{\pi}_{t+1} = \boldsymbol{\pi}_t P$.

Under mild conditions (irreducibility and aperiodicity), the chain admits a unique stationary distribution $\boldsymbol{\pi}^*$ satisfying $\boldsymbol{\pi}^* = \boldsymbol{\pi}^* P$. The stationary probability of being in the depressed state, π_D^* , gives the expected long-run fraction of time spent depressed.

Fitting. Transition probabilities are estimated from NEMESIS. The recovery probability P_{DR} is taken as the mean of the positive empirical hazard rates $\hat{h}_i$ across intervals. The irreversible fraction $S(\infty) \approx 0.20$ observed in the NEMESIS survival curve is used as the stationary depressed fraction, from which P_{RD} is recovered via the balance equation $\pi_D^* \cdot P_{DR} = \pi_R^* \cdot P_{RD}$. This yields:

$$P_{DD} = 1 - P_{DR} = 0.889, \quad P_{DR} = 0.111, \quad P_{RD} = 0.028, \quad P_{RR} = 0.972 \quad (3)$$

The resulting stationary distribution gives $\pi_D^* = 0.20$, consistent with the irreversible fraction.

4.1.2 Hidden Markov Model Extension

The standard Markov model requires hard thresholding of continuous scores into discrete states, which introduces misclassification error when scores fall near the boundary. Furthermore, self-report scores are noisy measurements of the true underlying state. The hidden Markov model (HMM) addresses both concerns by treating the true state S_t as latent and the observed hazard rate $\hat{h}_t$ as a noisy emission.

The HMM adds an emission model to the Markov transition structure:

$$\hat{h}_t \mid S_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2) \quad (4)$$

where μ_D and μ_R are the mean hazard rates associated with the high-recovery and low-recovery phases respectively, and σ is the emission noise controlling the confidence of state assignments. Rather than assigning each interval to a state by hard threshold, the HMM computes a posterior probability $P(S_t \mid \hat{h}_1, \dots, \hat{h}_T)$ for each interval.

Fitting. Parameters are estimated via the Baum-Welch algorithm, an expectation-maximization procedure applied to the sequence of 17 empirical hazard rates from NEMESIS. The algorithm alternates between computing posterior state probabilities and updating transition and emission parameters until convergence. Fitted emission means are $\hat{\mu}_D = 0.164$ and $\hat{\mu}_R = 0.028$.

The algorithm assigns the early high-hazard intervals, roughly months 0 through 7, to the high-recovery phase, and the later low-hazard tail, months 8 through 24, to the low-recovery phase. Clinically, this reflects a well-documented feature of MDD recovery. The bulk of recoveries occur within the first six months, after which the remaining cohort consists pre-

dominantly of chronic cases with low and roughly constant recovery rates. It is important to note that these hidden states describe phases of population recovery dynamics, not the states of any individual patient. The HMM observes a single sequence of 17 hazard rate values, one per inspection interval, and infers which phase the population as a whole occupied at each time point. The sensitivity analysis reveals the role of emission noise σ in driving this inference. At $\sigma = 0.02$, state boundaries are sharp; and intervals are assigned confidently to one phase or the other with posterior probabilities near 0 or 1. As σ increases to 0.20, the two emission distributions overlap heavily, state means converge, and most intervals receive uncertain assignments with posterior probabilities near 0.5. This illustrates the core advantage of the HMM over hard thresholding. Rather than forcing a binary classification, it quantifies how much the data actually supports a given state assignment.

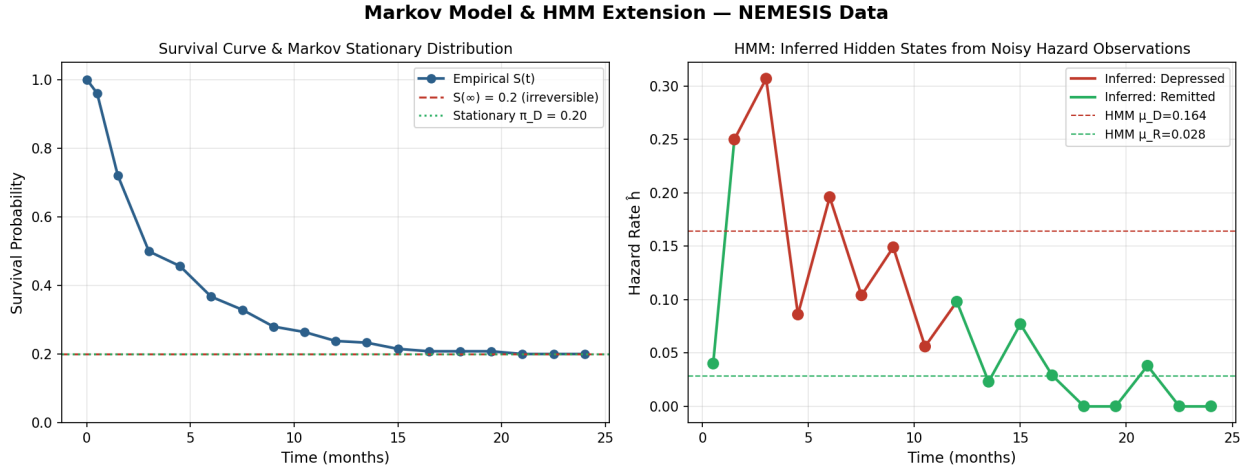


Figure 2: Left: Kaplan-Meier survival curve from NEMESIS with the irreversible fraction $S(\infty) = 0.20$ and stationary distribution $\pi_D^* = 0.20$ overlaid. Right: HMM inferred hidden states from the empirical hazard rate sequence. Red segments indicate intervals assigned to the high-recovery (depressed) phase; green segments indicate the low-recovery (remitted) phase.

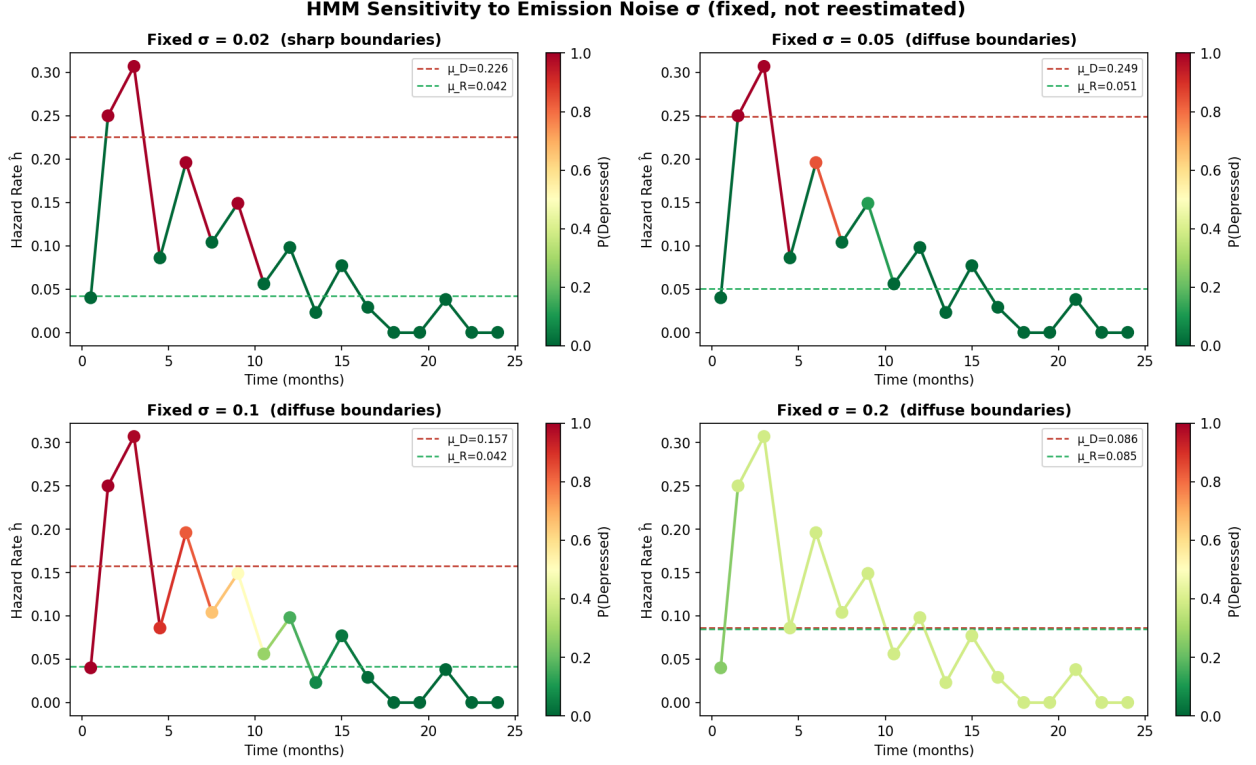


Figure 3: Sensitivity of HMM state assignments to fixed emission noise σ . Point color encodes the posterior probability of being in the high-recovery phase $P(\text{Depressed})$. At low σ (top left), state boundaries are sharp; at high σ (bottom right), assignments become diffuse and uncertain.

4.2 Autoregressive Model and Treatment Extension

4.2.1 AR(1) Gamma Model

The autoregressive model treats depression severity as a continuous time series, modeling each observation as a function of the previous one. Following [Cochran et al. \[2017\]](#), I adopt a first-order autoregressive model with Gamma-distributed observations, motivated by the non-negative, positively skewed nature of self-report scores. The model is:

$$X_{t+1} \mid X_t \sim \text{Gamma}(\text{mean} = \mu_t, \text{dispersion} = \alpha), \quad \mu_t = aX_t + b \quad (5)$$

The parameter $a \in (0, 1)$ is the persistence or “stickiness” of symptoms: high a means today’s severity strongly predicts tomorrow’s. The parameter $b > 0$ is the baseline drift, and the implied stationary mean is $b/(1 - a)$. The dispersion parameter α controls the noise level around the predicted mean.

Fitting via MLE. Parameters a , b , and α are estimated by maximizing the log-likelihood:

$$\ell(a, b, \alpha) = \sum_{t=1}^{T-1} \log p_{\Gamma}(X_{t+1}; \mu_t, \alpha) \quad (6)$$

where p_{Γ} denotes the Gamma probability density. Optimization is performed via the Nelder-Mead method on the 238-day Kossakowski daily mood series. Fitted values are $\hat{a} = 0.30$, $\hat{b} = 1.60$, implying a stationary mean of $1.60/(1 - 0.30) = 2.29$.

The low fitted persistence $\hat{a} = 0.30$ is notable, as it indicates that this individual’s daily affect is only weakly autocorrelated, with today’s mood score explaining approximately 30% of tomorrow’s expected score. This is consistent with the high-frequency ESM measurement context, where daily affect fluctuates rapidly compared to the slower timescales of clinical episode dynamics.

4.2.2 Treatment Extension

Effective treatment for depression does not merely reduce symptom severity at a single point in time. It changes the dynamics of the system: how sticky symptoms are, how quickly scores drift toward a long-run baseline, and what that baseline is. The AR(1) framework captures this naturally by allowing parameters to shift at a known treatment onset τ .

Two formulations are possible. The first adds an explicit treatment term:

$$\mu_t = aX_t + b + \beta u_t \quad (7)$$

where u_t is a treatment dosage or adherence variable and β measures its effect on the con-

ditional mean. This formulation is expressive but requires longitudinal treatment records, including dosage and adherence data, that are rarely available in observational datasets. The second formulation, a structural break, requires only the timing of treatment onset:

$$a \rightarrow a_{\text{pre}}, a_{\text{post}}, \quad b \rightarrow b_{\text{pre}}, b_{\text{post}} \quad (8)$$

Pre- and post-treatment parameters are estimated separately from the two segments of the time series. The structural break approach is implemented here because treatment timing is far more accessible in clinical practice than precise dosage records, and because splitting the estimation problem into two independent MLE problems is computationally straightforward.

No longitudinal dataset with known treatment timing was available for this analysis. Instead, I construct a synthetic demonstration to show what treatment-induced parameter shifts would look like under the AR(1) model. A 60-step time series is simulated in two segments. In the pre-treatment segment (steps 1–30), parameters are set to $a_{\text{pre}} = 0.88$ and $b_{\text{pre}} = 0.80$, representing high symptom stickiness and an elevated baseline. The implied stationary mean is $b_{\text{pre}}/(1 - a_{\text{pre}}) = 6.67$. At step 30, parameters shift to $a_{\text{post}} = 0.50$ and $b_{\text{post}} = 0.30$, representing reduced persistence and a lower baseline following treatment. The implied stationary mean falls to $b_{\text{post}}/(1 - a_{\text{post}}) = 0.60$, an 91% reduction. The simulated trajectory shows the severity score rising and fluctuating at high levels before treatment, then dropping sharply and stabilizing near the new lower stationary mean afterward. This demonstration isolates not just a reduction in severity at one time point, but a measurable change in the underlying generative parameters of the process. Applied to real clinical data with known treatment timing, the same approach would quantify how much a given intervention changed both the stickiness and the long-run level of a patient’s symptom trajectory.

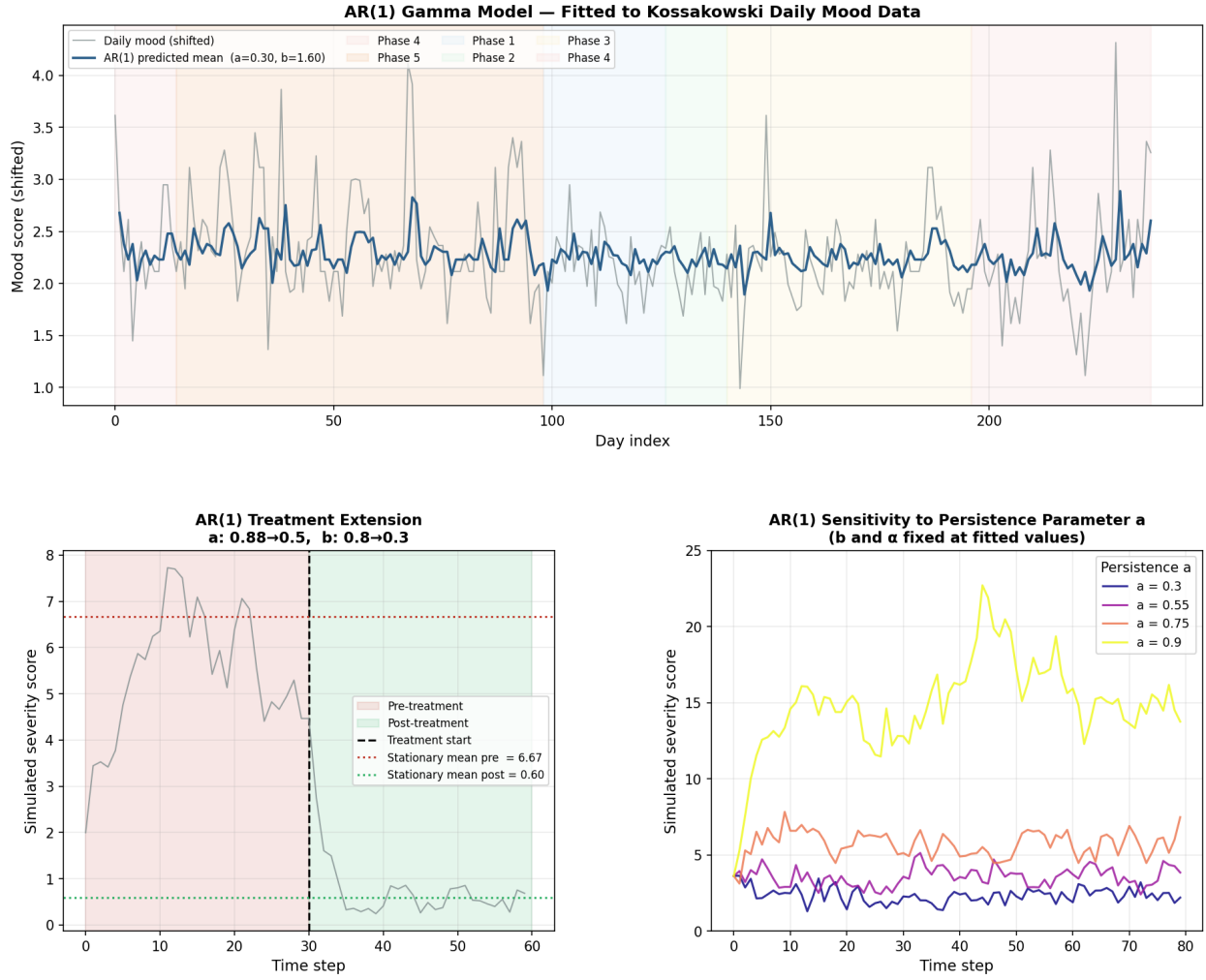


Figure 4: Top: AR(1) Gamma model fitted to 238 days of Kossakowski daily mood data. The blue line shows the one-step-ahead predicted mean; shading indicates study phases. Bottom left: simulated treatment extension with structural break at step 30, showing the drop in stationary mean from 6.67 to 0.60. Bottom right: sensitivity of simulated trajectories to the persistence parameter a with stationary mean fixed.

4.3 Ornstein-Uhlenbeck Random Mood Model and Individual Extension

4.3.1 OU Model Formulation

The Ornstein-Uhlenbeck (OU) random mood model, introduced by [van der Werf et al. \[2006\]](#), describes mood as a continuous latent process governed by:

$$\frac{dy}{dt} + \frac{1}{T} y = Z g_w(D) \quad (9)$$

where $y(t)$ is latent mood, T is the relaxation time (the timescale over which mood returns to baseline after a perturbation), Z is a susceptibility coefficient determining how strongly mood responds to stimuli, and $g_w(D)$ is intermittent Gaussian white noise with average stimulus spacing D . Specifically:

$$g_w(D) = \begin{cases} \mathcal{N}(0, 1) & \text{with probability } 1/D \\ 0 & \text{with probability } 1 - 1/D \end{cases} \quad (10)$$

This model is discretized for simulation with daily time steps $\Delta t = 1$:

$$y_n = y_{n-1} e^{-1/T} + Z \xi_n \quad (11)$$

A depressive episode is defined as $y(t)$ crossing below a threshold $\theta = -2$ and remaining there for at least 14 days, consistent with the DSM-5 minimum duration criterion.

Population fitting. To simulate a population, T is drawn from a Poisson distribution with mean $\langle T \rangle$ across subjects, representing individual heterogeneity in resilience. Episode durations from the simulation are binned into the same inspection intervals as NEMESIS, yielding simulated hazard rates $\hat{h}_i^{\text{sim}}$ that are compared to the empirical $\hat{h}_i$. The best-fitting parameters, found via grid search, are $\langle T \rangle = 365$ days and $D = 120$ days.

The OU simulation reproduces the empirical hazard rates reasonably well across the early intervals. From months 0 through 6, where NEMESIS data are richest, the simulated hazard curve tracks the shape of the empirical bars: low at the first interval, rising to a peak around months 3–4, then declining. Beyond month 10 the simulated hazard remains elevated at roughly 0.13–0.15 while the empirical bars drop toward zero, producing visible divergence. This is largely a data artifact rather than a model failure; by month 10 fewer than 51 subjects remain in the NEMESIS cohort, and by month 18 only 27 remain, with 6 censored. Hazard estimates computed from samples this small are highly variable, and several late intervals record zero recoveries simply because no one happened to recover during that window. The simulation, drawing on a cohort of 8,000 synthetic subjects, produces stable estimates throughout and should be compared primarily to the early intervals where the empirical signal is reliable. The epidemiological picture from NEMESIS is still clear. Approximately 75% of subjects recover within 12 months, and approximately 20% do not recover within the 24-month observation window, converging to the irreversible fraction $S(\infty) \approx 0.20$. Among those who do recover, the hazard rate is roughly constant across intervals, a pattern consistent with memoryless exponential decay and justifying the 1-step model of [van der Werf et al. \[2006\]](#). Depression, for the recoverable majority, does not become easier or harder to recover from the longer it has lasted.

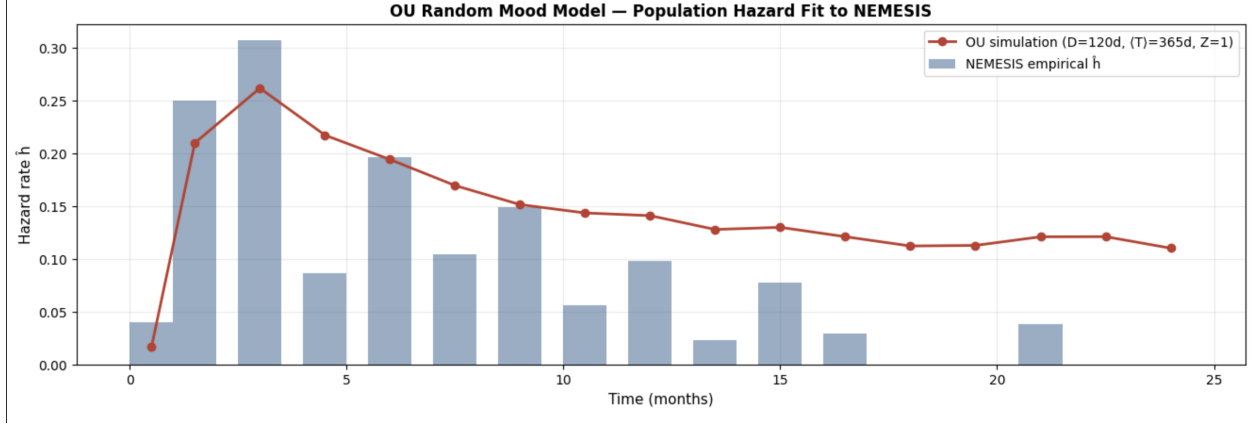


Figure 5: OU random mood model population hazard fit to NEMESIS data. Blue bars show empirical hazard rates $\hat{h}_i$; the red line shows simulated hazard rates with $\langle T \rangle = 365$ days, $D = 120$ days, $Z = 1$. Divergence at later intervals reflects sparse NEMESIS data beyond month 10.

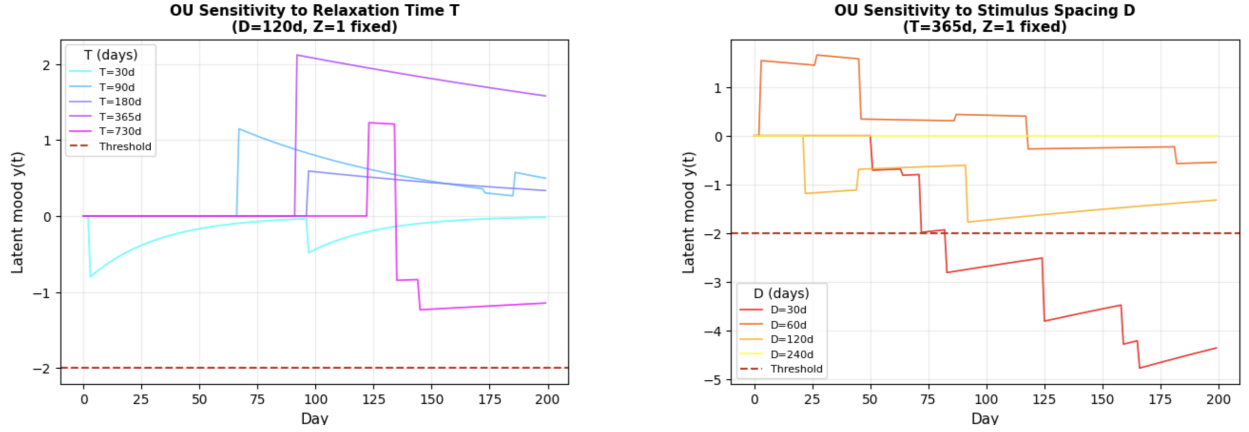


Figure 6: Sensitivity of OU latent mood trajectories to relaxation time T (left, $D = 120$ days fixed) and stimulus spacing D (right, $T = 365$ days fixed). Larger T produces slower mean reversion and longer excursions below the depression threshold. Smaller D increases stimulus frequency and produces more frequent threshold crossings.

4.3.2 Individual Extension via Kalman Filter

The population OU model describes average recovery dynamics across a cohort but says nothing about any individual's trajectory. To extend the model to the individual level, we introduce an observation model connecting the latent mood process to the observed daily

score:

$$X_t = y_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_v^2) \quad (12)$$

Combined with the discretized OU process (11), this constitutes a linear Gaussian state-space model:

$$y_t = \phi y_{t-1} + \sigma_w w_t, \quad w_t \sim \mathcal{N}(0, 1) \quad (13)$$

$$X_t = y_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_v^2) \quad (14)$$

where $\phi = e^{-1/T}$ and $\sigma_w = Z/\sqrt{D}$. For linear-Gaussian state-space models, the Kalman filter provides the exact optimal posterior estimate of the hidden state given observations.

The Kalman filter operates in two alternating steps at each time t :

$$\text{Predict: } \hat{y}_{t|t-1} = \phi \hat{y}_{t-1|t-1}, \quad P_{t|t-1} = \phi^2 P_{t-1|t-1} + \sigma_w^2 \quad (15)$$

$$\text{Update: } K_t = \frac{P_{t|t-1}}{P_{t|t-1} + \sigma_v^2}, \quad \hat{y}_{t|t} = \hat{y}_{t|t-1} + K_t(X_t - \hat{y}_{t|t-1}) \quad (16)$$

where K_t is the Kalman gain and $P_{t|t}$ is the posterior variance of the state estimate.

Parameters T , D , Z , and σ_v are estimated by maximizing the marginal log-likelihood of the observed sequence. Optimization over the 238-day Kossakowski series yields $\hat{T} = 7$ days, $\hat{D} = 44$ days, $\hat{Z} = 1.11$, $\hat{\sigma}_v = 0.42$.

The fitted $\hat{T} = 7$ days is dramatically shorter than the population estimate $\langle T \rangle = 365$ days. This contrast is itself a meaningful finding:

The gap between $\hat{T} = 7$ days and $\langle T \rangle = 365$ days reflects a difference in what each estimate measures, not a contradiction. The population T describes how long a full depressive episode takes to resolve at the clinical scale: months of sustained low mood before the system returns to baseline. The individual $\hat{T}$ describes how quickly daily affect recovers from moment-to-moment perturbations in ESM data. In essence, a bad afternoon does not predict a bad week.

The Kalman filter operates on daily mood fluctuations, so it naturally recovers the faster timescale. The filtered trajectory supports this, as latent mood $\hat{y}(t)$ stays well above the -2σ depression threshold throughout the observation period, consistent with a participant who is functionally stable during the early phases and shows increasing mood variability in the later phases without crossing into a clinically defined episode.

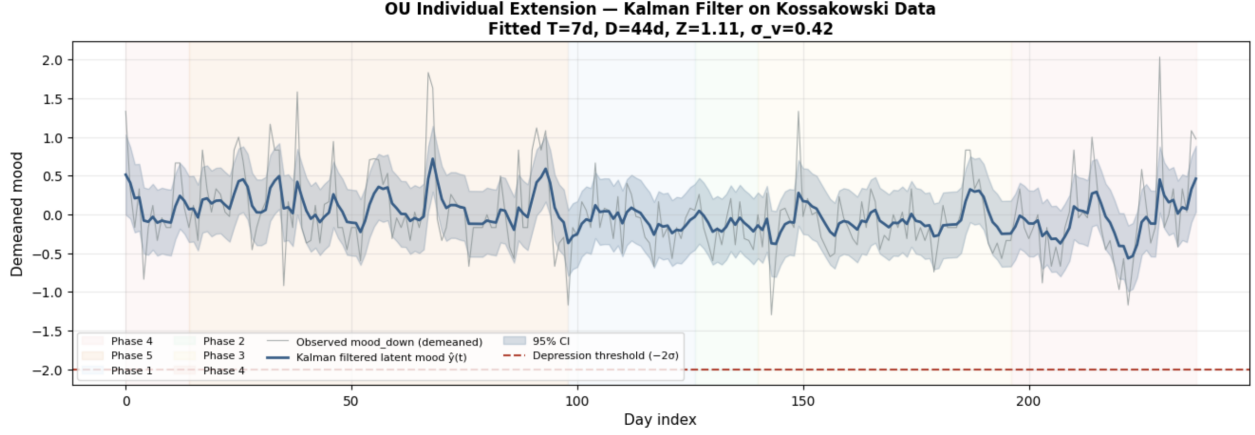


Figure 7: Kalman filter applied to 238 days of Kossakowski daily mood data. Gray line: raw observed mood scores (demeaned). Blue line: Kalman-filtered latent mood estimate $\hat{y}(t|t)$. Shaded band: 95% credible interval. Red dashed line: depression threshold at -2σ . Phase shading indicates the five study phases. Fitted parameters: $T = 7$ d, $D = 44$ d, $Z = 1.11$, $\sigma_v = 0.42$.

5 Synthesis and Model Comparison

The three models examined in this paper are not competing explanations of the same phenomenon. Instead, they operate at different levels of abstraction and address different clinical questions with the same underlying data.

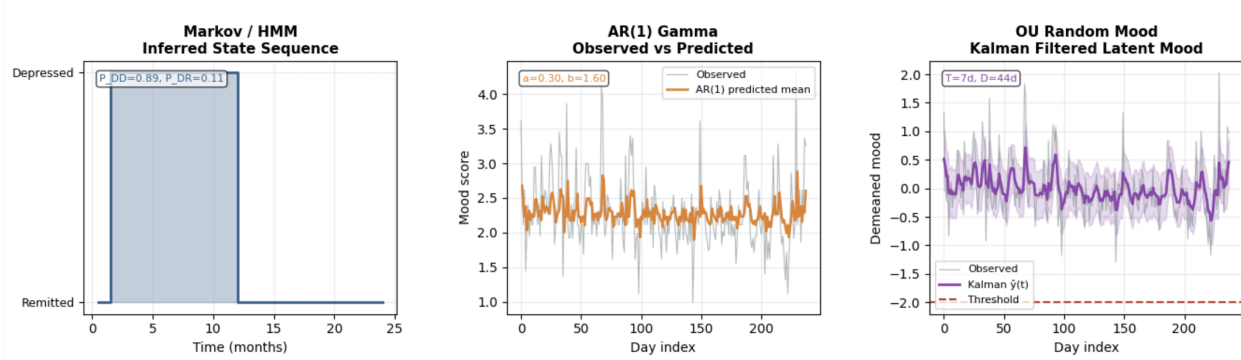


Figure 8: Three representations of the same Kossakowski individual data. Left: Markov/HMM inferred state sequence (NEMESIS-based transition probabilities applied to classify phases). Center: AR(1) observed versus one-step-ahead predicted mean. Right: OU Kalman-filtered latent mood trajectory with depression threshold.

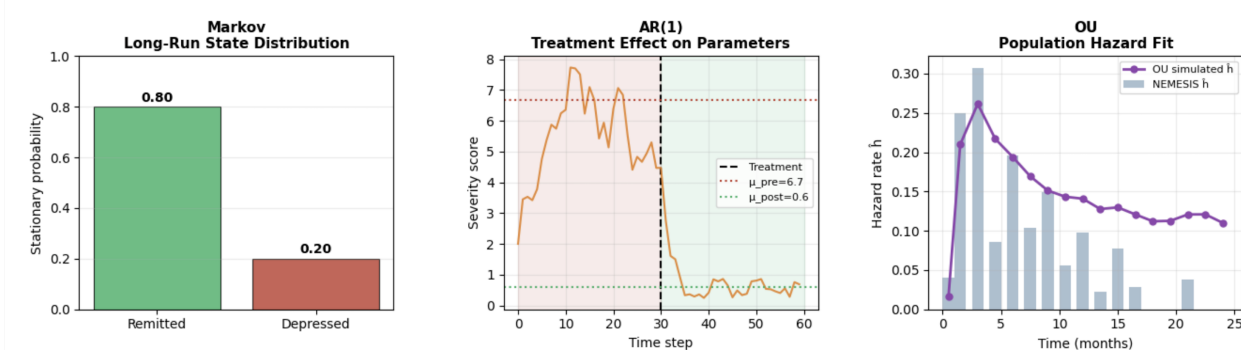


Figure 9: What each model tells you about the same data. Left: Markov long-run stationary distribution, with 80% of time in the remitted state and 20% in the depressed state. Center: AR(1) treatment extension simulation showing the drop in stationary severity mean from 6.67 to 0.60 at treatment onset. Right: OU population hazard fit to NEMESIS, showing the simulated recovery curve against empirical hazard rates.

Model Comparison Summary

	Markov / HMM	AR(1) Gamma	OU Random Mood
What it models	Discrete state transitions	Continuous severity over time	Latent mood + episode dynamics
Input data	Population episode counts	Individual weekly/daily scores	Population survival data
Key parameters	$P_{DD}, P_{DR}, P_{RD}, P_{RR}$	a (persistence), b (baseline)	T (resilience), D (stimulus spacing)
Extension	Hidden Markov Model	Treatment parameter shift	Individual Kalman filter
Best for	Relapse risk, time in state	Forecasting, treatment detection	Recovery dynamics, resilience
Main limitation	Memoryless, hard thresholds	Requires dense time series	Shocks unobserved, data-hungry

Figure 10: Summary comparison of the three modeling frameworks across key dimensions.

Figure 8 makes the complementarity of the three models concrete. The same 238 days of Kossakowski mood data look entirely different depending on which lens is applied. The Markov model reduces the record to a binary state sequence, collapsing continuous variation into a question of whether the population is in a high-recovery or low-recovery phase. The AR(1) model treats every daily value as signal, fitting a smooth predicted mean that tracks the data’s week-to-week drift. The Kalman filter strips out observation noise to recover a latent trajectory, widening its uncertainty bands as the later phases become more volatile. Each representation discards information the others preserve. None is complete alone.

The three models answer different clinical questions and require different data to do so. The Markov model is best suited to population-level questions about relapse risk and expected time spent in a depressed state. It requires only episode-level counts at regular intervals, making it applicable to administrative or epidemiological data where individual scores are unavailable. The AR(1) model addresses week-to-week severity dynamics - given today’s score, what is tomorrow’s expected score, and did a treatment intervention shift the underlying parameters? It requires a dense individual time series, ideally weekly or more frequent, over a period long enough to estimate both pre- and post-treatment regimes. The OU model operates at two scales simultaneously. Fitted to population survival data it characterizes average recovery timescales; fitted to individual daily scores via the Kalman filter it recovers personal resilience parameters T , D , and Z that have direct clinical interpretability.

The extensions improve on the base models in targeted ways. The HMM replaces the Markov model’s hard state thresholds with probabilistic inference, which matters most when scores cluster near the boundary between states. Rather than forcing an ambiguous interval into the depressed or remitted category, the HMM assigns it a posterior probability and propagates that uncertainty forward. The structural break extension gives the AR(1) a mechanism for detecting whether treatment changed the generative process, not just the level of a single observation. The Kalman filter extension moves the OU model from a population

summary to an individual inference tool, describing a specific patient’s resilience rather than the average patient’s. Figure 9 illustrates what each model concretely produces. The Markov stationary distribution quantifies long-run time in each state, the AR(1) treatment simulation shows how a structural break manifests in a simulated severity trajectory, and the OU hazard fit shows the population recovery curve the model is calibrated against.

On the other hand, Figure 10 puts the comparison into practical terms. A clinician managing relapse prevention at the population level would reach for the Markov or HMM framework, using fitted transition probabilities to estimate the probability of a patient remaining depressed across successive intervals. A clinician tracking an individual patient’s response to a new medication would use the AR(1) structural break to test whether persistence and baseline shifted after treatment onset. A researcher characterizing individual differences in resilience across a cohort would apply the Kalman filter OU extension, comparing fitted T values across patients to identify who recovers slowly and who does not. To restate this once more - the models are not alternatives to each other. They are tools for different questions, and the most complete picture of depression dynamics requires all three.

6 Discussion

The central result of this paper is not any single parameter estimate but a methodological one. Self-report severity scores, collected frequently enough, carry sufficient statistical structure to support three fundamentally different modeling frameworks simultaneously. The Markov and HMM analysis extracted phase structure from the shape of population hazard rates. The AR(1) extracted autocorrelation dynamics from daily mood fluctuations. The Kalman filter extracted a latent trajectory and noise decomposition from the same daily series. Each framework imposed different assumptions and asked different questions, yet all three found coherent signal in data that a skeptic might dismiss as too noisy or too subjective to model. The distributional structure of self-report scores, positively skewed, autocorrelated, mean-

reverting, is not incidental. It is the phenomenon.

The contrast between the NEMESIS parameters and the Kossakowski parameters is the most striking empirical finding. The population relaxation time $\langle T \rangle = 365$ days and the individual relaxation time $\hat{T} = 7$ days differ by a factor of 50. As stated before, these estimates do not contradict each other because they measure different things. The population T describes how long a clinical depressive episode persists before the cohort as a whole begins recovering. The individual $\hat{T}$ describes how quickly a single person’s daily affect rebounds from a momentary perturbation. A bad day is not a depressive episode. The Kossakowski participant’s mood fluctuates rapidly around a stable mean, bouncing back from daily lows within a week, while population-level episode resolution takes months. This scale separation suggests that daily ESM data and population survival data are not interchangeable inputs to the same model. They describe different layers of the same underlying process, and a complete account of depression dynamics requires both.

Three limitations of this analysis are important to recognize. The treatment extension in the AR(1) section is a synthetic demonstration, not an empirical result. The parameter values $a_{\text{pre}} = 0.88$ and $a_{\text{post}} = 0.50$ were chosen to produce a visually clear illustration of what a structural break looks like, not estimated from data with known treatment timing. I do not make any claim about what real antidepressant treatment does to AR(1) parameters. The HMM application is also unconventional, as the algorithm was applied to a sequence of 17 population-level hazard rates rather than an individual’s score trajectory, which is the setting for which HMMs are typically designed. The hidden states in this analysis represent population recovery phases, not patient states, and the model has far fewer observations than HMM estimation ordinarily requires. Finally, the Kalman filter’s fitted $\hat{T} = 7$ days sits at the lower bound imposed during optimization. The optimizer consistently pushed toward shorter relaxation times, suggesting the true maximum likelihood estimate may lie below 7 days. The bound was imposed on clinical grounds, but it means the fitted value should be

interpreted as a constrained estimate rather than a free optimum.

With richer data, each model would be substantially more informative. A dataset containing weekly PHQ-9 scores from multiple patients, half of whom began antidepressant treatment at a known date, would allow the AR(1) structural break to be estimated from real pre- and post-treatment segments rather than simulated. Comparing $\hat{a}_{\text{pre}}$ and $\hat{a}_{\text{post}}$ across patients would test whether treatment reliably reduces symptom persistence or whether the effect is heterogeneous. Applying the Kalman filter OU extension to multiple individuals would allow direct comparison of fitted T and Z values, turning the resilience parameters into individual traits rather than single estimates. A patient with $T = 14$ days recovers twice as fast from mood perturbations as one with $T = 28$ days, and that difference, measured from the first few weeks of daily tracking, might inform treatment intensity or follow-up scheduling before a full episode develops.

The broader implication is that personalized psychiatry does not require biological markers. The parameters T , Z , a , and b are all estimable from self-report time series that patients can generate themselves with a daily questionnaire. A clinician who knows a patient’s relaxation time has a quantitative description of their resilience. A clinician who can test whether $\hat{a}$ shifted after treatment onset has an objective signal of pharmacological response that does not depend on a patient’s ability to summarize their own improvement. None of this requires a blood draw or a brain scan. It requires consistent measurement over time and a model that is able to account for that measurement seriously.

Acknowledgments

I would like to thank Professor Reed for guidance and feedback throughout the development of this project!

I also used Claude (Anthropic, claude.ai) to assist with Python code development, LaTeX

typesetting (particularly for hyperlinking figures/references and writing mathematical formulas/symbols), and drafting the general skeletal structure as given in the table of contents. All prose writing, mathematical formulations, interpretations, and analytical decisions are my own.

References

- American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*. American Psychiatric Publishing, Washington, DC, 5th edition, 2013.
- Amy L. Cochran, André Schultz, Melvin G. McInnis, and Daniel B. Forger. A comparison of mathematical models of mood in bipolar disorder. In *Computational Neurology and Psychiatry*, volume 6 of *Springer Series in Bio-/Neuroinformatics*, pages 315–341. Springer, Cham, 2017. doi: 10.1007/978-3-319-49959-8_11.
- Jolanda J. Kossakowski, Peter C. Groot, Jonas M. B. Haslbeck, Denny Borsboom, and Marieke Wichers. Data from “Critical slowing down as a personalized early warning signal for depression”. *Journal of Open Psychology Data*, 5(1), 2017. doi: 10.5334/jopd.29.
- Kurt Kroenke, Robert L. Spitzer, and Janet B. W. Williams. The PHQ-9: Validity of a brief depression severity measure. *Journal of General Internal Medicine*, 16(9):606–613, 2001. doi: 10.1046/j.1525-1497.2001.016009606.x.
- Siebre Y. van der Werf, Kirsten I. Kaptein, Peter de Jonge, Jan Spijker, Ron de Graaf, and Jakob Korf. Major depressive episodes and random mood. *Archives of General Psychiatry*, 63(5):509–518, 2006. doi: 10.1001/archpsyc.63.5.509.